APM348 Mathematical Modelling – Midterm 1 (Group)

University of Toronto
Faculty of Arts & Science

Oct 6 2026
40 Minutes

Group Member #1

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Solution

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Exam Reminders:

- Fill out the name, student number, and email address for each group member at the top of this page.
- Do not begin writing the exam until instructed to do so.
- As a student, you help create a fair and inclusive writing environment; unauthorized aids are prohibited and using one may result in you being charged with an academic offense.
- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. These devices may not be left in your pockets.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator.
- In the event of a fire alarm, do not check cell phones or other electronic devices unless authorized to do so.

Special Instructions:

- Write legibly and darkly.
- For questions with a boxed area, ensure your answer is completely within the box.
- Fill in your bubbles completely.
- This booklet contains 6 pages. If your booklet is missing a page, please raise your hand to notify an exam facilitator.
- **Do not remove or unstaple any page from this booklet.**

Scoring

Question:	**1**	**Total**
Points:	10	**10**

For the questions below, you can use the Jupyter Notebook:

`https://uoft.me/apm348-f26-midterm1-group`

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1 Suppose a handblown wine glass factory receives an order for G wine glasses to be delivered at time T.

The glasses are fragile, so storing them is costly, so they want to don't want to produce them early. On the other hand, producing the glasses late incurs a larger cost because producing glasses faster is more wasteful.

Assume that initially we don't have any glasses stored.

Wwe define
- $t =$ the amount of time (in days) since the order was placed;
- $x(t) =$ the (accumulated) number of wine glasses **produced** by time t for this order.

We want a better model than the one we used in the individual part of the test.

The production cost includes the fact that producing glasses faster costs more:

$$P(x) = \int_0^T c\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 \mathrm{d}t$$

(a) (3 points) The cost for storing the glasses should include the following:
- A fraction p of all glasses in storage brake every day, so they must be produced again;

To consider this fact, define the following:
- $I(t) =$ the number of wine glasses **in storage** at time t for this order.

Observe that because of the glasses broken, $x(t) \neq I(t)$.

Write $\frac{\mathrm{d}I}{\mathrm{d}t}$ in terms of $I(t)$ and $\frac{\mathrm{d}x}{\mathrm{d}t}$. Also, what is $I(0)$ and $I(T)$? Explain your answers.

We have the following meaning for these quantities:
- $\frac{dI}{dt} \approx$ daily change of the number of glasses in storage
- $I \approx$ number of glasses in storage
- $\frac{dx}{dt} \approx$ daily number of glasses produced

So change of glasses in storage = glasses produced − glasses broken

$\frac{\mathrm{d}I}{\mathrm{d}t} =$ $\frac{dx}{dt} - pI$

$I(0) =$ 0 we start without any glasses in storage

$I(T) =$ G we want to have G glasses at time T (only unbroken ones)

(b) (2 points) We obtain the following optimization problem:

$$\min c \int_0^T \underbrace{\left(\frac{\mathrm{d}I}{\mathrm{d}t}(t) + pI(t)\right)^2}_{G(t,I,I')} \mathrm{d}t$$

where $I(0)$ and $I(T)$ satisfies the conditions you found in the previous part.

Use the Euler-Lagrange equation to find a differential equation and constraints that the optimal production rate $x(t)$ must satisfy. Justify your answer.

$I(t)$

The Euler-Lagrange equation yields: $\frac{d}{dt}\left(\frac{\partial G}{\partial I'}\right) = \frac{\partial G}{\partial I}$.

$\frac{\partial G}{\partial I'} = 2(I' + pI) \Rightarrow \frac{d}{dt}\left(\frac{\partial G}{\partial I'}\right) = 2I'' + 2pI'$

$\frac{\partial G}{\partial I} = 2(I' + pI)p$

So we get $2I'' + 2pI' = 2pI' + 2p^2 I$

$I'' = p^2 I$

ODE:

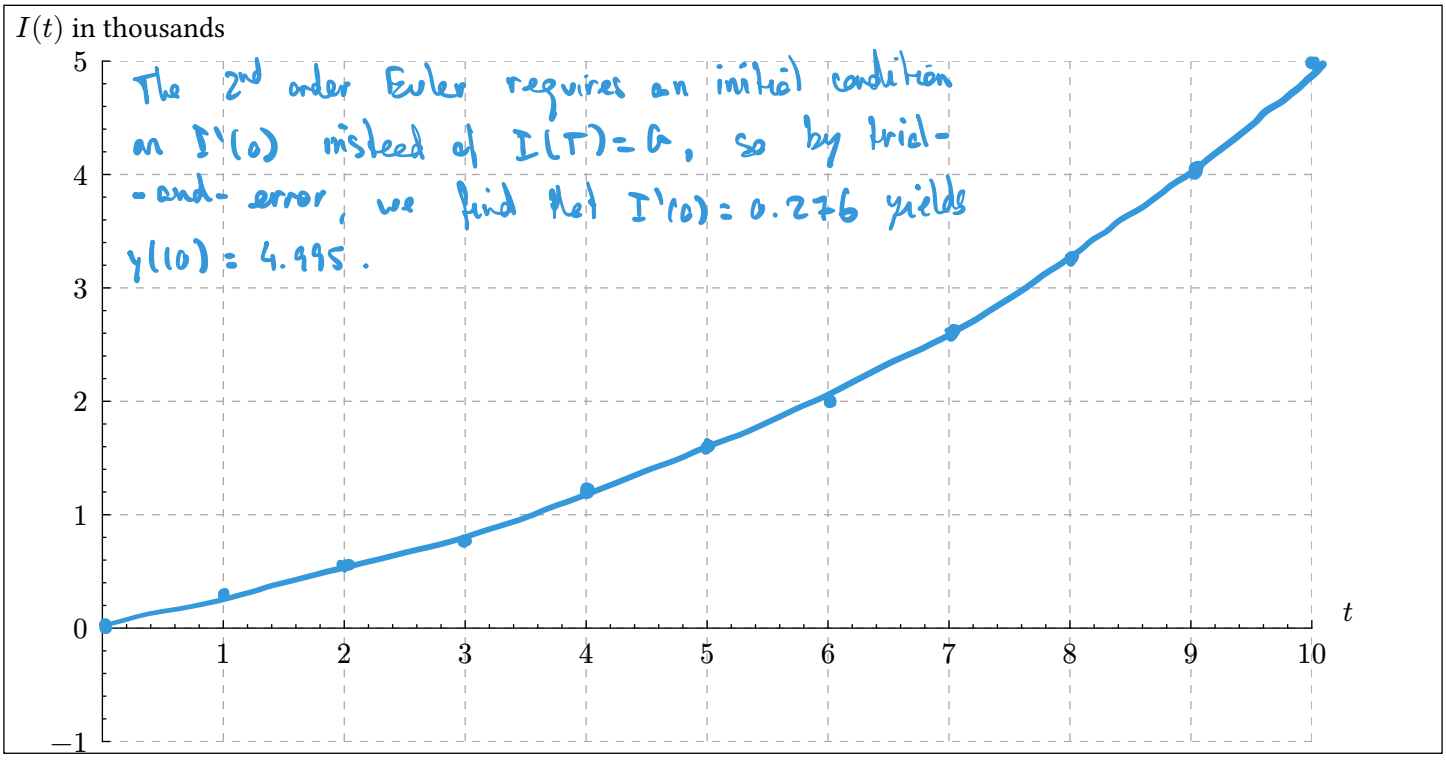
$I'' = p^2 I$

$I(0) = 0$

$I(T) = G$

(c) (3 points) We want to produce 5000 glasses in 10 days, with $p = 0.2$, and $c = 1$.

Numerically solve the differential equation you found in the previous part and sketch the optimal production rate $I(t)$.



(d) (2 points) With the same parameters as in the previous part, use a Riemann sum to estimate the total production cost of the optimal production rate you found in the previous part

$$\int_0^T \left(\frac{\mathrm{d}I}{\mathrm{d}t}(t) + pI(t)\right)^2 \mathrm{d}t.$$

Explain how you calculated the cost.

We take the variable $y \approx I(t)$ and $z \approx \frac{dI}{dt}$ and calculate the Riemann sum $\sum_{i=0}^{n-1} h \cdot (z_i + p \cdot y_i)^2$

where $h = 0.01$ and $n = 1000$.

Cost of optimal production = 10.215

(e) *(2 bonus points)* With the same parameters as in the previous parts, estimate the total number of broken glasses. Explain your calculations.

Approximately, the factory breaks $pI(t)$ glasses per day, so the total glasses broken can be estimated as $\int_0^{10} pI(t)\,dt$.

We use the Riemann sum to approximate it:

$$\sum_{i=0}^{n-1} h \cdot p \cdot y_i$$

Total number of broken glasses = 3.815

USE THIS PAGE FOR SCRATCH PAPER OR TO CONTINUE OTHER QUESTIONS.

If you wish to have this page marked, make sure to refer to it in your original solution.

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